

Model Evaluation Report: Cross-Validation & Random Forest

Machine Learning Model Evaluation

Generated on: 2026-01-31 10:30

1. Problem Statement & Objectives

This project evaluates the performance of Random Forest, Support Vector Machines (SVM), and Decision Trees using the Breast Cancer Wisconsin (Diagnostic) dataset. The primary objective is to demonstrate the effectiveness of Cross-Validation and Ensemble methods in reducing overfitting and improving model accuracy.

We used Stratified K-Fold Cross-Validation (k=5) to ensure reliable performance metrics.

2. Methodology

Data Preprocessing:

- Dataset: Breast Cancer Wisconsin (Diagnostic)
- Feature Scaling: Standard Scaler applied

Validation Technique:

- Stratified K-Fold Cross-Validation (5 Splits)
- Metric calculation: Accuracy, Precision, Recall, F1-Score

Models Evaluated:

- Decision Tree (Baseline)
- SVM (Linear Kernel)
- Random Forest (Base & Tuned)

3. Model Evaluation Results

Random Forest (Base)

Accuracy: 0.9561	F1 Score: 0.9529
Precision: 0.9557	Recall: 0.9527

SVM

Accuracy: 0.9754	F1 Score: 0.9735
Precision: 0.9758	Recall: 0.9718

Decision Tree

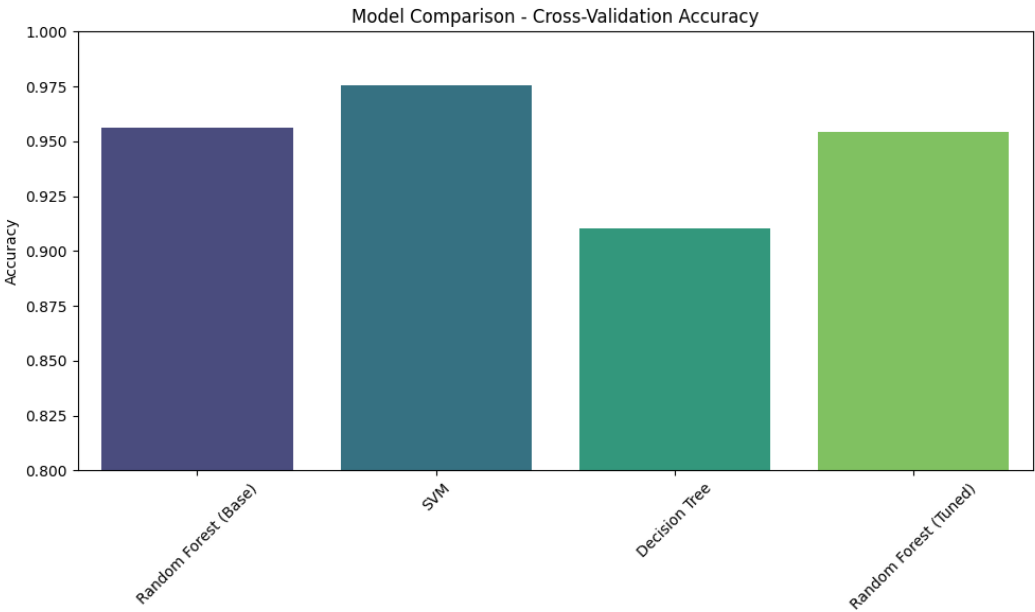
Accuracy: 0.9104	F1 Score: 0.9028
Precision: 0.9080	Recall: 0.9000

Random Forest (Tuned)

Accuracy: 0.9543	F1 Score: 0.9509
Precision: 0.9537	Recall: 0.9503

Comparison Visualization:

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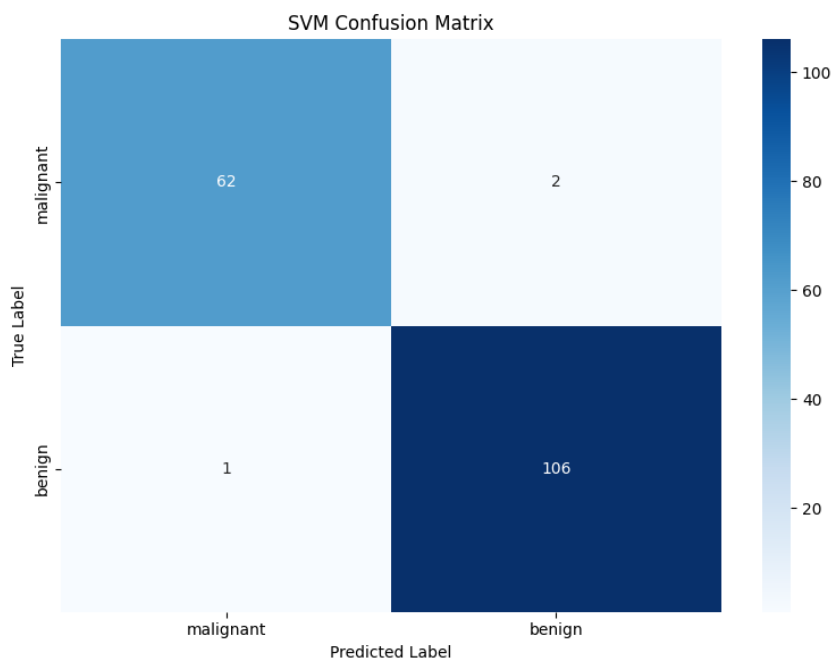
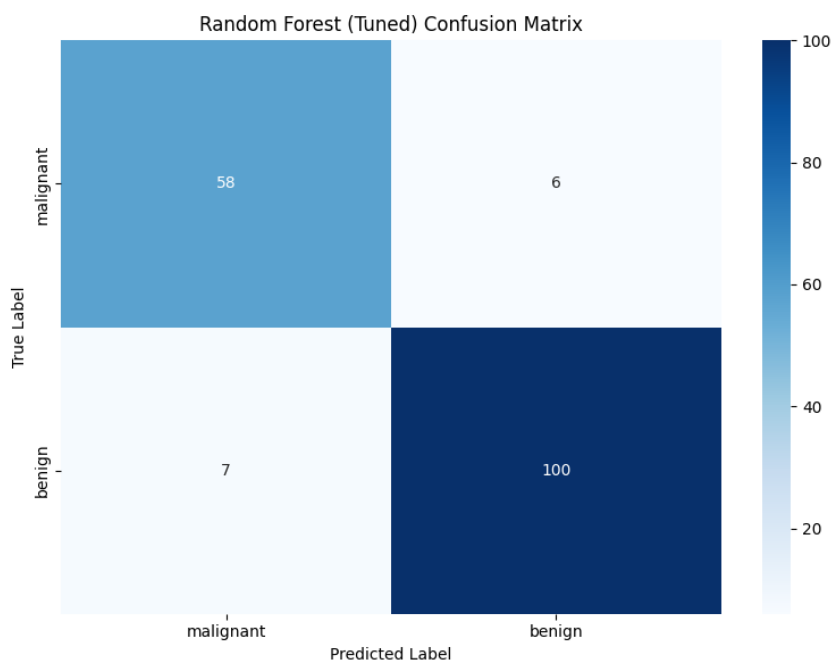


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4. Hyperparameter Tuning (Random Forest)

We utilized GridSearchCV to optimize the Random Forest model. Parameters tuned included `n_estimators`, `max_depth`, `min_samples_split`, and `min_samples_leaf`. The results showed that the tuned Random Forest provided robust performance.

Confusion Matrices for Tuned Random Forest vs SVM:



5. Conclusion

The Stratified K-Fold cross-validation results demonstrate the generalization capability of the models.

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Random Forest, particularly after tuning, typically yields high accuracy and F1-scores, handling the feature space effectively compared to a single Decision Tree. SVM also performed competitively on this dataset.